

Data

ImageCAS-X test split
160 CT angiography scans
left and right trees
distance-map radius

Lesion insertion

cosine stenosis, 40-80% DS
LAD, LCx, RCA; 10/20 mm
6 944 eligible instances
150 in six FFR bands

Segmentation error

T1 missed side branch
T2 vessel break
T3 lesion +2.46 mm
T4 taper $\times 0.930$

Boundary conditions

A fixed (clean bed)
B re-derived on
corrupted tree
C one scaling fitted to
clean territory flows

Reduced-order FFR

leaky bed: 150
discrete bed: 97
steady hyperaemia
FFR 20 mm distal

Outcomes

Δ FFR vs clean model
flip at FFR 0.80
residual $< 10\%$ and
 $|\Delta$ FFR| > 0.05

3D case study (one LAD lesion, 80% DS): baseline and T1 lumens solved in OpenFOAM under fixed resistances and prescribed territory flows, beside reduced-order twins